

```
In [1]: import pandas as pd
import seaborn as sns

%matplotlib inline
import matplotlib.pyplot as plt
```

```
In [2]: file_path = r"D:\Ignou, Code\TusharTemp\Python Numpy and Pandas\Jupyter Lab\Projects\Data\TWO_CENTURIES_OF_UM_RACES.csv"

df = pd.read_csv(file_path)
```

C:\Users\Tushar\AppData\Local\Temp\ipykernel_6740\2937814502.py:3: DtypeWarning: Columns (11) have mixed types. Specify dtype option on import or set low_memory=False.

```
df = pd.read_csv(file_path)
```

```
In [3]: df.head(10)
```

Out[3]:

	Year of event	Event dates	Event name	Event distance/ length	Event number of finishers	Athlete performance	Athlete club	Athlete country	Athlete year of birth	Athlete gender	Athlete age category	Athlete average speed	Athlete ID
0	2018	06.01.2018	Selva Costera (CHI)	50km	22	4:51:39 h	Tnfrc	CHI	1978.0	M	M35	10.286	0
1	2018	06.01.2018	Selva Costera (CHI)	50km	22	5:15:45 h	Roberto Echeverría	CHI	1981.0	M	M35	9.501	1
2	2018	06.01.2018	Selva Costera (CHI)	50km	22	5:16:44 h	Puro Trail Osorno	CHI	1987.0	M	M23	9.472	2
3	2018	06.01.2018	Selva Costera (CHI)	50km	22	5:34:13 h	Columbia	ARG	1976.0	M	M40	8.976	3
4	2018	06.01.2018	Selva Costera (CHI)	50km	22	5:54:14 h	Baguales Trail	CHI	1992.0	M	M23	8.469	4
5	2018	06.01.2018	Selva Costera (CHI)	50km	22	6:25:01 h	NaN	ARG	1974.0	M	M40	7.792	5
6	2018	06.01.2018	Selva Costera (CHI)	50km	22	6:28:00 h	Los Patagones	ARG	1979.0	F	W35	7.732	6
7	2018	06.01.2018	Selva Costera (CHI)	50km	22	6:32:24 h	Reactiva Chile	CHI	1967.0	F	W50	7.645	7
8	2018	06.01.2018	Selva Costera (CHI)	50km	22	6:39:08 h	Puro Trail Osorno	CHI	1985.0	M	M23	7.516	8

	Year of event	Event dates	Event name	Event distance/ length	Event number of finishers	Athlete performance	Athlete club	Athlete country	Athlete year of birth	Athlete gender	Athlete age category	Athlete average speed	Athlete ID
9	2018	06.01.2018	Selva Costera (CHI)	50km	22	6:45:11 h	Marlene Flores Team	CHI	1976.0	M	M40	7.404	9

Checking the datatypes of each columns

In [4]: `df.dtypes`

```
Out[4]: Year of event          int64
Event dates          object
Event name           object
Event distance/length object
Event number of finishers int64
Athlete performance  object
Athlete club         object
Athlete country      object
Athlete year of birth float64
Athlete gender       object
Athlete age category object
Athlete average speed object
Athlete ID           int64
dtype: object
```

Total rows and columns

In [5]: `df.shape`

```
Out[5]: (7117634, 13)
```

Extracting only races where Country = USA AND Year = 2020 AND Distance = 50mi/50km

Data with 50km

```
In [6]: fifty_km_race = df[df['Event distance/length'] == '50km']  
fifty_km_race.shape
```

```
Out[6]: (1503475, 13)
```

Data with 50mi

```
In [7]: fifty_mi_race = df[df['Event distance/length'] == '50mi']  
fifty_mi_race.shape
```

```
Out[7]: (333685, 13)
```

Data with only 50km and 50mi races

```
In [8]: specified_distance = ['50km', '50mi']  
distance_filter = df['Event distance/length'].isin(specified_distance)  
  
distance_filter.shape
```

```
Out[8]: (7117634,)
```

Filtering rows with only USA event races

```
In [9]: # Making a new column that have only the country name from Event Names  
df['Country Names Only'] = df['Event name'].str.extract(r'\((.*?)\)')  
df.head(10)
```

Out[9]:

	Year of event	Event dates	Event name	Event distance/ length	Event number of finishers	Athlete performance	Athlete club	Athlete country	Athlete year of birth	Athlete gender	Athlete age category	Athlete average speed	Athlete ID	Countr Name Onl
0	2018	06.01.2018	Selva Costera (CHI)	50km	22	4:51:39 h	Tnfr	CHI	1978.0	M	M35	10.286	0	Cl
1	2018	06.01.2018	Selva Costera (CHI)	50km	22	5:15:45 h	Roberto Echeverría	CHI	1981.0	M	M35	9.501	1	Cl
2	2018	06.01.2018	Selva Costera (CHI)	50km	22	5:16:44 h	Puro Trail Osorno	CHI	1987.0	M	M23	9.472	2	Cl
3	2018	06.01.2018	Selva Costera (CHI)	50km	22	5:34:13 h	Columbia	ARG	1976.0	M	M40	8.976	3	Cl
4	2018	06.01.2018	Selva Costera (CHI)	50km	22	5:54:14 h	Baguales Trail	CHI	1992.0	M	M23	8.469	4	Cl
5	2018	06.01.2018	Selva Costera (CHI)	50km	22	6:25:01 h	NaN	ARG	1974.0	M	M40	7.792	5	Cl
6	2018	06.01.2018	Selva Costera (CHI)	50km	22	6:28:00 h	Los Patagones	ARG	1979.0	F	W35	7.732	6	Cl
7	2018	06.01.2018	Selva Costera (CHI)	50km	22	6:32:24 h	Reaktiva Chile	CHI	1967.0	F	W50	7.645	7	Cl
8	2018	06.01.2018	Selva Costera (CHI)	50km	22	6:39:08 h	Puro Trail Osorno	CHI	1985.0	M	M23	7.516	8	Cl

	Year of event	Event dates	Event name	Event distance/ length	Event number of finishers	Athlete performance	Athlete club	Athlete country	Athlete year of birth	Athlete gender	Athlete age category	Athlete average speed	Athlete ID	Countr Name Only
9	2018	06.01.2018	Selva Costera (CHI)	50km	22	6:45:11 h	Marlene Flores Team	CHI	1976.0	M	M40	7.404	9	Ch

```
In [10]: # Filtering dataframe rows where country name is USA
country_filter = df[df['Country Names Only'] == 'USA']

country_filter.shape
```

```
Out[10]: (1355619, 14)
```

```
In [11]: # Country filter in pandas series
country_filter = df['Country Names Only'] == 'USA'
```

Filtering year to 2020

```
In [12]: year_filter = df[df['Year of event'] == 2020]
year_filter.head()
```

Out[12]:

	Year of event	Event dates	Event name	Event distance/ length	Event number of finishers	Athlete performance	Athlete club	Athlete country	Athlete year of birth	Athlete gender	Athlete age category	Athlete average speed	Athlete ID
2538401	2020	07.-08.02.2020	Taipei 48hr Ultra Marathon - 24h Split (TPE)	24h	38	219.453 km	日本隊	JPN	1965.0	M	M50	9.144	5316
2538402	2020	07.-08.02.2020	Taipei 48hr Ultra Marathon - 24h Split (TPE)	24h	38	202.878 km	NaN	TPE	1964.0	M	M55	8.453	446
2538403	2020	07.-08.02.2020	Taipei 48hr Ultra Marathon - 24h Split (TPE)	24h	38	194.259 km	台灣大 腳丫長 跑協會	TPE	1969.0	F	W50	8.094	6396
2538404	2020	07.-08.02.2020	Taipei 48hr Ultra Marathon - 24h Split (TPE)	24h	38	190.944 km	NaN	TPE	1968.0	M	M50	7.956	1746
2538405	2020	07.-08.02.2020	Taipei 48hr Ultra Marathon	24h	38	186.303 km	日本隊	JPN	1960.0	M	M60	7.763	5066

Year of event	Event dates	Event name	Event distance/length	Event number of finishers	Athlete performance	Athlete club	Athlete country	Athlete year of birth	Athlete gender	Athlete age category	Athlete average speed	Athlete
		- 24h Split (TPE)										

```
In [13]: # Year filter in pandas series
year_filter = df['Year of event'] == 2020
```

Combining all the filters (Distance filter, Country filter, and Year Filter) to get the required result

```
In [14]: df2 = df[country_filter & distance_filter & year_filter]
df2.shape
```

Out[14]: (26090, 14)

Rechecking the result where Country = USA AND Year = 2020 AND Distance = 50mi/50km

```
In [15]: event_country = df2['Country Names Only'].unique()
print(f"Event Country: {event_country}")

year_of_event = df2['Year of event'].unique()
print(f"Year of Event: {year_of_event} ")

event_distance = df2['Event distance/length'].unique()
print(f"Event Distance: {event_distance} ")
```

Event Country: ['USA']

Year of Event: [2020]

Event Distance: ['50km' '50mi']

```
In [16]: # Dropping the temporary column 'Country Names Only' because it is redundant and of no use
df2.drop(columns=['Country Names Only'], inplace=True)
df2.head()
```



```
C:\Users\Tushar\AppData\Local\Temp\ipykernel_6740\3267217944.py:2: SettingWithCopyWarning:  
A value is trying to be set on a copy of a slice from a DataFrame
```

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
df2.drop(columns=['Country Names Only'], inplace=True)
```

Out[16]:

	Year of event	Event dates	Event name	Event distance/ length	Event number of finishers	Athlete performance	Athlete club	Athlete country	Athlete year of birth	Athlete gender	Athlete age category	Athlete average speed	Athlete ID
2539945	2020	02.02.2020	West Seattle Beach Run - Winter Edition (USA)	50km	20	3:17:55 h	*Normandy Park, WA	USA	1991.0	M	M23	15.158	71287
2539946	2020	02.02.2020	West Seattle Beach Run - Winter Edition (USA)	50km	20	4:02:32 h	*Gold Bar, WA	USA	1981.0	M	M35	12.369	629508
2539947	2020	02.02.2020	West Seattle Beach Run - Winter Edition (USA)	50km	20	4:07:57 h	*Vashon, WA	USA	1999.0	M	MU23	12.099	64838
2539948	2020	02.02.2020	West Seattle Beach Run - Winter Edition (USA)	50km	20	4:22:02 h	*Gig Harbor, WA	USA	1983.0	M	M35	11.449	704450
2539949	2020	02.02.2020	West Seattle Beach Run -	50km	20	4:27:34 h	*Bainbridge Island, WA	USA	1977.0	M	M40	11.212	810281

Year of event	Event dates	Event name	Event distance/ length	Event number of finishers	Athlete performance	Athlete club	Athlete country	Athlete year of birth	Athlete gender	Athlete age category	Athlete average speed	Athlete ID
---------------------	----------------	---------------	------------------------------	------------------------------------	------------------------	-----------------	--------------------	-----------------------------	-------------------	----------------------------	-----------------------------	---------------

Winter
Edition
(USA)

```
In [17]: df2.shape
```

```
Out[17]: (26090, 13)
```

Removing '(USA)' from Event Dates column because all the data is of USA

```
In [18]: df2['Event name'] = df2['Event name'].str.replace(r'\((.*?)\)', '', regex=True)
df2.head(10)
```

C:\Users\Tushar\AppData\Local\Temp\ipykernel_6740\2714549301.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
df2['Event name'] = df2['Event name'].str.replace(r'\((.*?)\)', '', regex=True)

Out[18]:

	Year of event	Event dates	Event name	Event distance/ length	Event number of finishers	Athlete performance	Athlete club	Athlete country	Athlete year of birth	Athlete gender	Athlete age category	Athlete average speed	Athlet II
2539945	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	3:17:55 h	*Normandy Park, WA	USA	1991.0	M	M23	15.158	7128
2539946	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:02:32 h	*Gold Bar, WA	USA	1981.0	M	M35	12.369	62950
2539947	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:07:57 h	*Vashon, WA	USA	1999.0	M	MU23	12.099	6483
2539948	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:22:02 h	*Gig Harbor, WA	USA	1983.0	M	M35	11.449	70445
2539949	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:27:34 h	*Bainbridge Island, WA	USA	1977.0	M	M40	11.212	81028

	Year of event	Event dates	Event name	Event distance/ length	Event number of finishers	Athlete performance	Athlete club	Athlete country	Athlete year of birth	Athlete gender	Athlete age category	Athlete average speed	Athlet II
2539950	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:42:06 h	*Seattle, WA	USA	1985.0	F	W23	10.635	81028
2539951	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:49:20 h	*Camano Island, WA	USA	1961.0	M	M55	10.369	1173
2539952	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:49:50 h	*Clinton, WA	USA	1970.0	M	M45	10.351	8039
2539953	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:51:00 h	*Seattle, WA	USA	1975.0	F	W40	10.309	14090
2539954	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	5:02:35 h	*Sammamish, WA	USA	1979.0	M	M40	9.915	75388

Creating athlete age column (2020 - Athlete year of birth)

```
In [19]: df2["Athlete Age"] = 2020 - df2['Athlete year of birth']  
df2.head()
```

C:\Users\Tushar\AppData\Local\Temp\ipykernel_6740\10434534.py:1: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.

Try using `.loc[row_indexer,col_indexer] = value` instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
df2["Athlete Age"] = 2020 - df2['Athlete year of birth']
```

Out[19]:

	Year of event	Event dates	Event name	Event distance/ length	Event number of finishers	Athlete performance	Athlete club	Athlete country	Athlete year of birth	Athlete gender	Athlete age category	Athlete average speed	Athlete ID
2539945	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	3:17:55 h	*Normandy Park, WA	USA	1991.0	M	M23	15.158	71287
2539946	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:02:32 h	*Gold Bar, WA	USA	1981.0	M	M35	12.369	629508
2539947	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:07:57 h	*Vashon, WA	USA	1999.0	M	MU23	12.099	64838
2539948	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:22:02 h	*Gig Harbor, WA	USA	1983.0	M	M35	11.449	704450
2539949	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:27:34 h	*Bainbridge Island, WA	USA	1977.0	M	M40	11.212	810281

Removing 'h' from athlete performance column

```
In [20]: df2['Athlete performance'] = df2['Athlete performance'].str.replace(r'\s*h$', '', regex=True)
df2.head()
# \s* matches optional spaces before h

# h matches the letter h

# $ ensures that h is at the end of the string
```

C:\Users\Tushar\AppData\Local\Temp\ipykernel_6740\20796798.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
df2['Athlete performance'] = df2['Athlete performance'].str.replace(r'\s*h$', '', regex=True)
```


Out[20]:

	Year of event	Event dates	Event name	Event distance/ length	Event number of finishers	Athlete performance	Athlete club	Athlete country	Athlete year of birth	Athlete gender	Athlete age category	Athlete average speed	Athlete ID
2539945	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	3:17:55	*Normandy Park, WA	USA	1991.0	M	M23	15.158	71287
2539946	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:02:32	*Gold Bar, WA	USA	1981.0	M	M35	12.369	629508
2539947	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:07:57	*Vashon, WA	USA	1999.0	M	MU23	12.099	64838
2539948	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:22:02	*Gig Harbor, WA	USA	1983.0	M	M35	11.449	704450
2539949	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:27:34	*Bainbridge Island, WA	USA	1977.0	M	M40	11.212	810281

Drop columns: Athlete club, Athlete Country, Athlete Age Category, Athlete Year of birth

```
In [21]: columns_to_drop = ['Athlete club', 'Athlete age category', 'Athlete year of birth', 'Athlete country']
df2.drop(columns=columns_to_drop, inplace=True)

df2.shape
```

C:\Users\Tushar\AppData\Local\Temp\ipykernel_6740\3891889345.py:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
df2.drop(columns=columns_to_drop, inplace=True)
```

```
Out[21]: (26090, 10)
```

```
In [22]: df2.head()
```

Out[22]:

	Year of event	Event dates	Event name	Event distance/length	Event number of finishers	Athlete performance	Athlete gender	Athlete average speed	Athlete ID	Athlete Age
2539945	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	3:17:55	M	15.158	71287	29.0
2539946	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:02:32	M	12.369	629508	39.0
2539947	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:07:57	M	12.099	64838	21.0
2539948	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:22:02	M	11.449	704450	37.0
2539949	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:27:34	M	11.212	810281	43.0

In [23]: `list(df2.columns)`

```
Out[23]: ['Year of event',  
          'Event dates',  
          'Event name',  
          'Event distance/length',  
          'Event number of finishers',  
          'Athlete performance',  
          'Athlete gender',  
          'Athlete average speed',  
          'Athlete ID',  
          'Athlete Age']
```

Cleaning up null values

```
In [24]: df2.isna().sum()
```

```
Out[24]: Year of event          0
Event dates                    0
Event name                     0
Event distance/length          0
Event number of finishers      0
Athlete performance            0
Athlete gender                 0
Athlete average speed          0
Athlete ID                     0
Athlete Age                    233
dtype: int64
```

```
In [25]: df2 = df2.dropna()
```

```
In [26]: df2.isna().sum()
```

```
Out[26]: Year of event          0
Event dates                    0
Event name                     0
Event distance/length          0
Event number of finishers      0
Athlete performance            0
Athlete gender                 0
Athlete average speed          0
Athlete ID                     0
Athlete Age                    0
dtype: int64
```

```
In [27]: df2.shape
```

```
Out[27]: (25857, 10)
```

Checking for duplicate values

```
In [28]: df2[df2.duplicated()]
```

Out[28]:

Year of event	Event dates	Event name	Event distance/ length	Event number of finishers	Athlete performance	Athlete gender	Athlete average speed	Athlete ID	Athlete Age
------------------	----------------	---------------	---------------------------	------------------------------	------------------------	-------------------	--------------------------	---------------	----------------

Resetting index

In [29]:

```
df2.reset_index(drop=True, inplace=True)  
df2.head(10)
```

Out[29]:

	Year of event	Event dates	Event name	Event distance/length	Event number of finishers	Athlete performance	Athlete gender	Athlete average speed	Athlete ID	Athlete Age
0	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	3:17:55	M	15.158	71287	29.0
1	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:02:32	M	12.369	629508	39.0
2	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:07:57	M	12.099	64838	21.0
3	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:22:02	M	11.449	704450	37.0
4	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:27:34	M	11.212	810281	43.0
5	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:42:06	F	10.635	810282	35.0
6	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:49:20	M	10.369	11739	59.0
7	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:49:50	M	10.351	80394	50.0
8	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:51:00	F	10.309	140909	45.0
9	2020	02.02.2020	West Seattle Beach Run -	50km	20	5:02:35	M	9.915	753889	41.0

Year of event	Event dates	Event name	Event distance/length	Event number of finishers	Athlete performance	Athlete gender	Athlete average speed	Athlete ID	Athlete Age
		Winter Edition							

Fixing Datatypes

```
In [30]: df2.dtypes
```

```
Out[30]: Year of event      int64
Event dates      object
Event name       object
Event distance/length  object
Event number of finishers  int64
Athlete performance  object
Athlete gender      object
Athlete average speed  object
Athlete ID          int64
Athlete Age         float64
dtype: object
```

```
In [31]: # Athlete age to integer
df2['Athlete Age'] = df2['Athlete Age'].astype(int)
```

```
In [32]: # Athlete average speed to float
df2['Athlete average speed'] = df2['Athlete average speed'].astype(float)
```

```
In [33]: df2.head()
```

Out[33]:

	Year of event	Event dates	Event name	Event distance/length	Event number of finishers	Athlete performance	Athlete gender	Athlete average speed	Athlete ID	Athlete Age
0	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	3:17:55	M	15.158	71287	29
1	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:02:32	M	12.369	629508	39
2	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:07:57	M	12.099	64838	21
3	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:22:02	M	11.449	704450	37
4	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:27:34	M	11.212	810281	43

Renaming columns

```
In [34]: df2 = df2.rename(columns={
    'Year of event' : 'year',
    'Event dates' : 'race_day',
    'Event name' : 'race_name',
    'Event distance/length' : 'race_length',
    'Event number of finishers' : 'race_number_of_finishers',
    'Athlete performance' : 'athlete_performance',
    'Athlete gender' : 'athlete_gender',
    'Athlete average speed' : 'athlete_average_speed',
    'Athlete ID' : 'athlete_id',
    'Athlete Age' : 'athlete_age'
})
```



```
In [35]: df2.head()
```

```
Out[35]:
```

	year	race_day	race_name	race_length	race_number_of_finishers	athlete_performance	athlete_gender	athlete_average_speed	athlete_id
0	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	3:17:55	M	15.158	710
1	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:02:32	M	12.369	629
2	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:07:57	M	12.099	640
3	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:22:02	M	11.449	704
4	2020	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	4:27:34	M	11.212	810

Rearranging columns

```
In [36]: df3 = df2[['race_day', 'race_name', 'race_length', 'race_number_of_finishers', 'athlete_id', 'athlete_gender', 'athlete_age'],  
df3.head()
```

Out[36]:

	race_day	race_name	race_length	race_number_of_finishers	athlete_id	athlete_gender	athlete_age	athlete_performance	athlete_average
0	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	71287	M	29	3:17:55	
1	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	629508	M	39	4:02:32	
2	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	64838	M	21	4:07:57	
3	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	704450	M	37	4:22:02	
4	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	810281	M	43	4:27:34	

In [37]:

```
df3[df3['athlete_id'] == 222509]
```

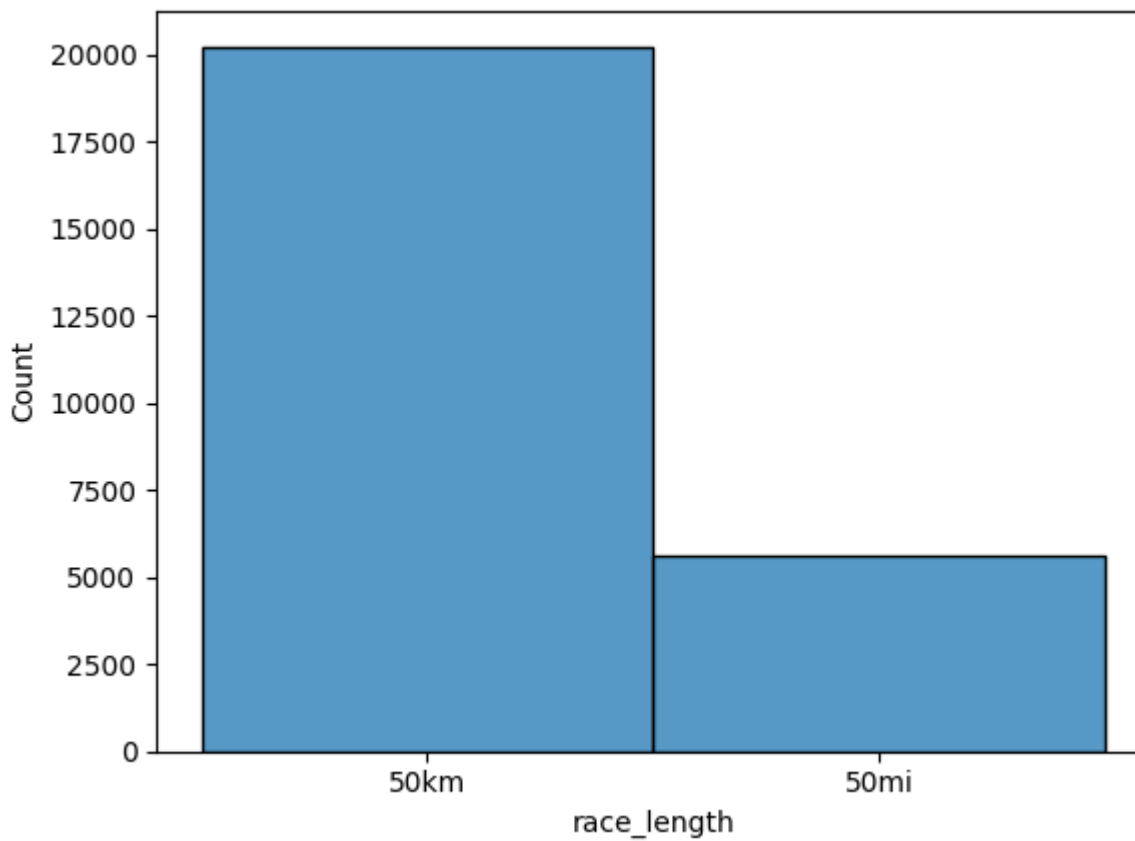
```
Out[37]:
```

	race_day	race_name	race_length	race_number_of_finishers	athlete_id	athlete_gender	athlete_age	athlete_performance	athlete_
7087	14.03.2020	Everglades 50 Mile Ultra Run	50mi	40	222509	M	23	9:19:10	
11054	22.02.2020	Manasota Track Club 50K	50km	36	222509	M	23	4:49:16	

Graphs and Plots

```
In [38]: # 50km vs 50mi  
sns.histplot(df3['race_length'])
```

```
Out[38]: <Axes: xlabel='race_length', ylabel='Count'>
```



Women in 50mi vs 50km

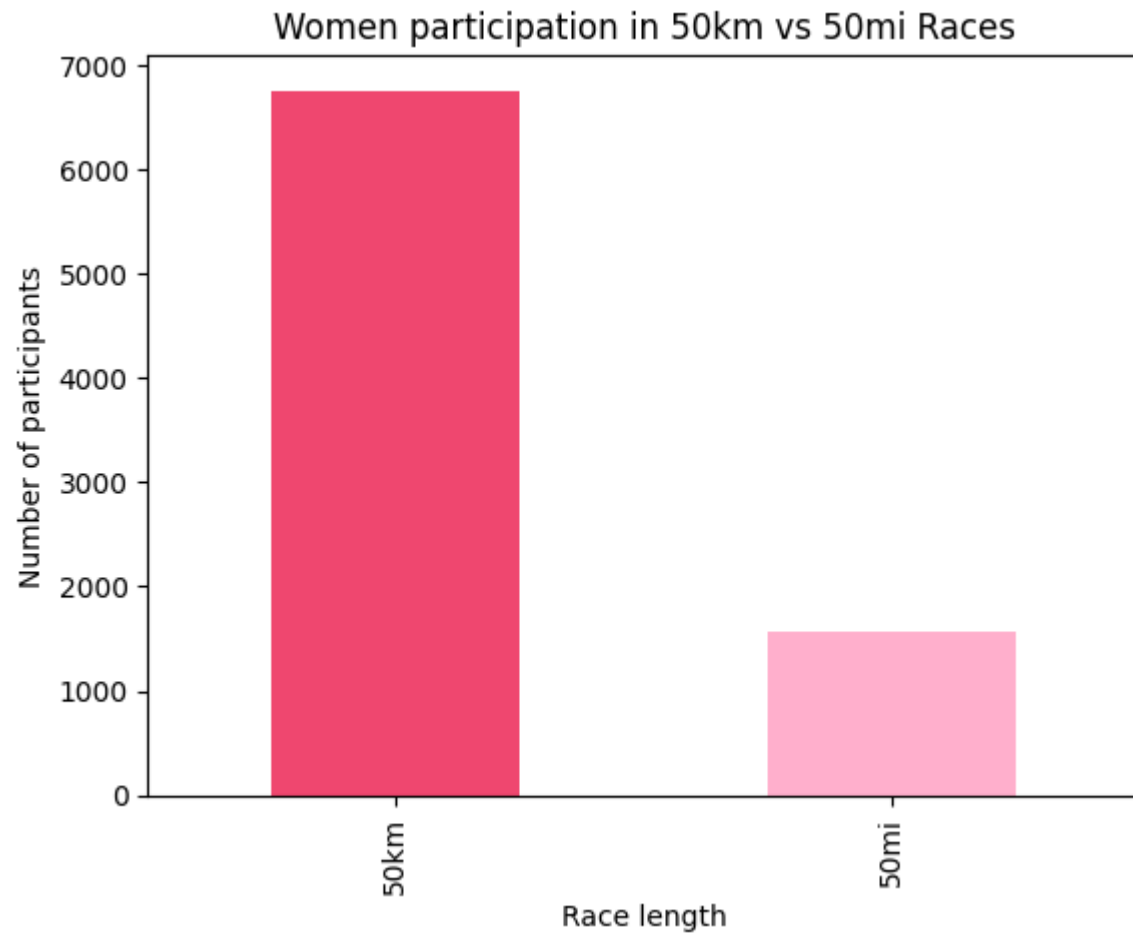
```
In [39]: women_races = df3[df3['athlete_gender'] == 'F']  
women_races['race_length'].value_counts()
```

```
Out[39]: race_length  
50km      6761  
50mi      1576  
Name: count, dtype: int64
```

```
In [40]: # race counts  
only_women = df3[df3['athlete_gender'] == 'F']
```

```
race_count = only_women['race_length'].value_counts()
```

```
In [41]: race_count.plot(kind='bar', color=['#ef476f', '#ffa6cc'])  
plt.title('Women participation in 50km vs 50mi Races')  
plt.ylabel('Number of participants')  
plt.xlabel('Race length')  
plt.grid(False)  
plt.show()
```



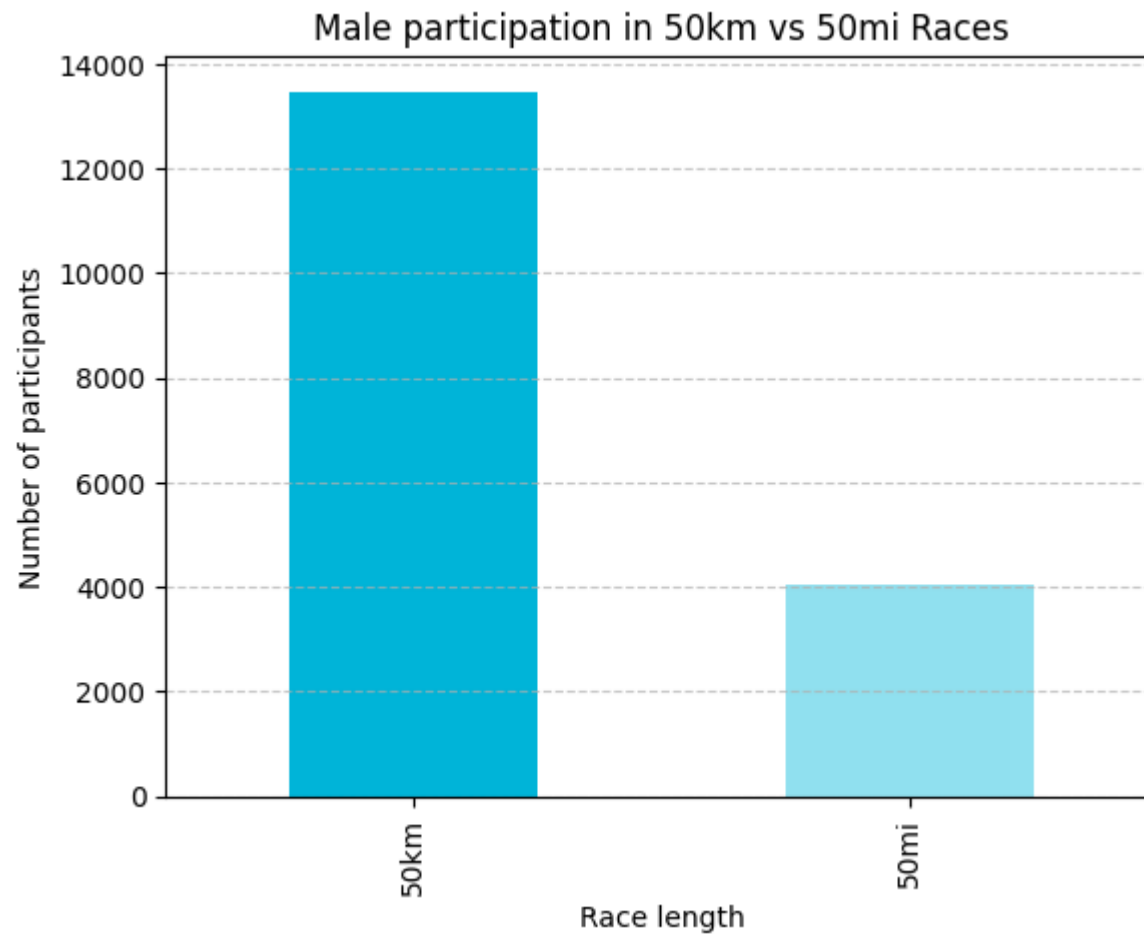
Male in 50mi vs 50km

```
In [42]: only_male = df3[df3['athlete_gender'] == 'M']  
only_male['race_length'].value_counts()
```

```
Out[42]: race_length  
50km      13469  
50mi       4051  
Name: count, dtype: int64
```

```
In [43]: # Race count  
race_count_male = only_male['race_length'].value_counts()
```

```
In [44]: race_count_male.plot(kind='bar', color=['#00b4d8', '#90e0ef']) # x-axis will be the index of the series (50mi and 50km). y-axis  
plt.title('Male participation in 50km vs 50mi Races')  
plt.xlabel('Race length')  
plt.ylabel('Number of participants')  
plt.grid(axis='y', linestyle='--', alpha=0.7)  
plt.show()
```



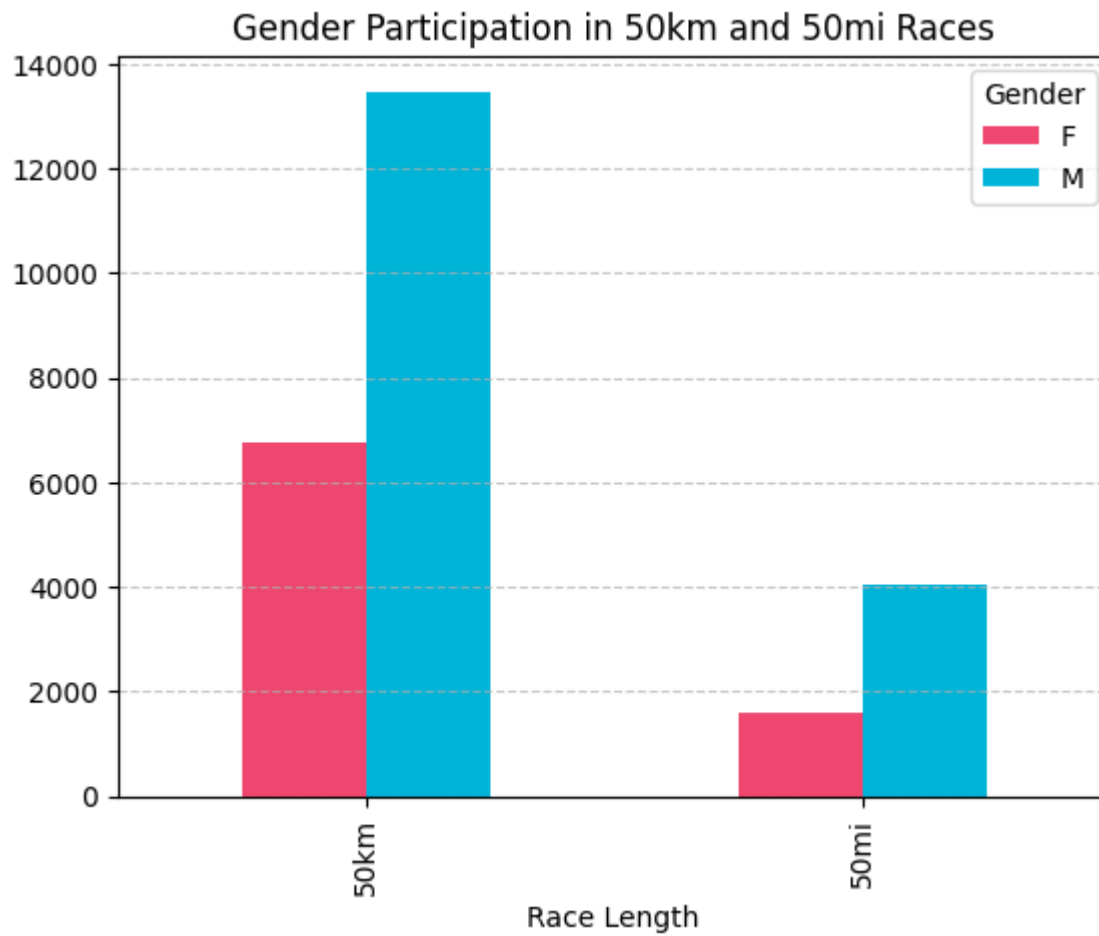
Male vs Female participation

```
In [45]: # Counting how many participants fall in each group
gender_race_count = df3.groupby(['race_length', 'athlete_gender']).size().unstack(fill_value=0)
gender_race_count
```

```
Out[45]:
```

athlete_gender	F	M
race_length		
50km	6761	13469
50mi	1576	4051

```
In [46]: gender_race_count.plot(kind='bar', color=['#ef476f', '#00b4d8'])
plt.title('Gender Participation in 50km and 50mi Races')
plt.xlabel('Race Length')
plt.legend(title='Gender')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```

Difference in speed for 50mi, 50km, male to female

```
In [47]: df3.groupby(['race_length', 'athlete_gender'], as_index=False)['athlete_average_speed'].mean()
```

```
Out[47]:
```

	race_length	athlete_gender	athlete_average_speed
0	50km	F	7.083011
1	50km	M	7.738985
2	50mi	F	6.834371
3	50mi	M	7.257633

Average age participation male to female

```
In [48]: df3.groupby(['race_length', 'athlete_gender'], as_index=False)['athlete_age'].mean()
```

```
Out[48]:
```

	race_length	athlete_gender	athlete_age
0	50km	F	42.027067
1	50km	M	42.205286
2	50mi	F	40.656726
3	50mi	M	41.051839

Season for data (Checking if people were slower in summer than winters)

```
In [49]: df3['race_month'] = df3['race_day'].str.split('.').str.get(1).astype(int)
df3.head(10)
```

Out[49]:

	race_day	race_name	race_length	race_number_of_finishers	athlete_id	athlete_gender	athlete_age	athlete_performance	athlete_average
0	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	71287	M	29	3:17:55	
1	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	629508	M	39	4:02:32	
2	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	64838	M	21	4:07:57	
3	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	704450	M	37	4:22:02	
4	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	810281	M	43	4:27:34	
5	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	810282	F	35	4:42:06	
6	02.02.2020	West Seattle Beach Run	50km	20	11739	M	59	4:49:20	

	race_day	race_name	race_length	race_number_of_finishers	athlete_id	athlete_gender	athlete_age	athlete_performance	athlete_average
		- Winter Edition							
		West Seattle							
7	02.02.2020	Beach Run - Winter Edition	50km	20	80394	M	50	4:49:50	
		West Seattle							
8	02.02.2020	Beach Run - Winter Edition	50km	20	140909	F	45	4:51:00	
		West Seattle							
9	02.02.2020	Beach Run - Winter Edition	50km	20	753889	M	41	5:02:35	

```
In [50]: df3['race_season'] = df3['race_month'].apply(
    lambda x: 'Winter' if x in [12, 1, 2] else
              'Spring' if x in [3, 4, 5] else
              'Summer' if x in [6, 7, 8] else
              'Autumn'
)

df3.head()
```

Out[50]:

	race_day	race_name	race_length	race_number_of_finishers	athlete_id	athlete_gender	athlete_age	athlete_performance	athlete_average_speed
0	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	71287	M	29	3:17:55	18.05
1	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	629508	M	39	4:02:32	16.85
2	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	64838	M	21	4:07:57	16.65
3	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	704450	M	37	4:22:02	16.05
4	02.02.2020	West Seattle Beach Run - Winter Edition	50km	20	810281	M	43	4:27:34	15.85

In [51]:

```
# Cheking season-wise average speed of athlete in 50mi races  
df3.query('race_length == "50mi"]').groupby(['race_season'], as_index=False)['athlete_average_speed'].mean().sort_values('athlete_average_speed')
```

```
Out[51]:
```

	race_season	athlete_average_speed
3	Winter	7.303244
0	Autumn	7.270117
1	Spring	7.082557
2	Summer	6.505776

```
In [52]: # Athletes were fastest in winter and slowest in summer (50mi races)
```

```
In [53]: # Cheking season-wise average speed of athlete in 50km races
season_wise_50km_races = df3.query('race_length == "50km"').groupby(['race_season'], as_index=False)['athlete_average_speed'].
season_wise_50km_races
```

```
Out[53]:
```

	race_season	athlete_average_speed
1	Spring	7.889135
3	Winter	7.653627
0	Autumn	7.342964
2	Summer	7.031116

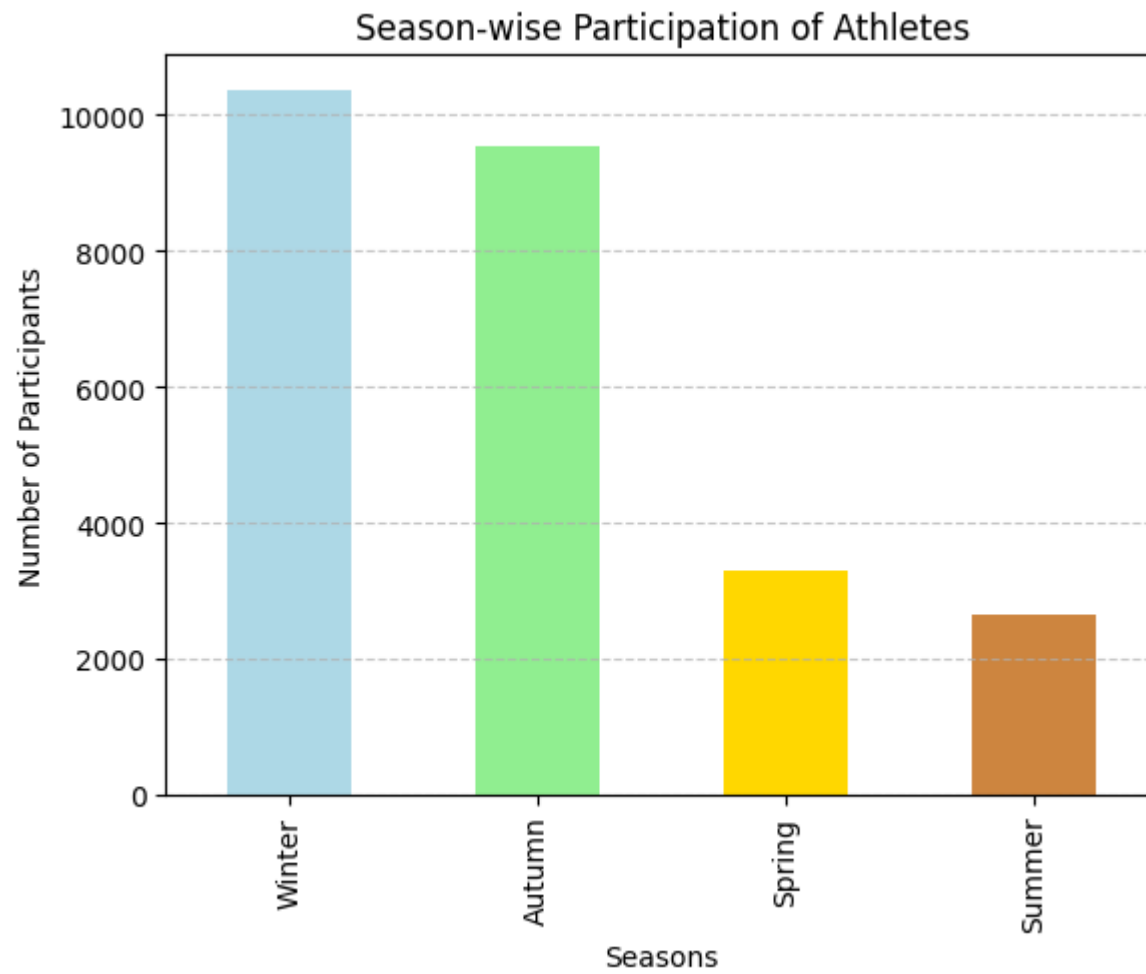
```
In [54]: # Athletes were fastest in spring and slowest in summer (50km races)
```

Season-wise participation of athletes

```
In [55]: # Season-wise participation of athletes
season_counts = df3['race_season'].value_counts()
season_counts
```

```
Out[55]: race_season
        Winter    10361
        Autumn     9549
        Spring     3294
        Summer     2653
        Name: count, dtype: int64
```

```
In [56]: season_counts.plot(kind='bar', color=['#ADD8E6', '#90EE90', '#FFD700', '#CD853F'])
plt.title('Season-wise Participation of Athletes')
plt.xlabel('Seasons')
plt.ylabel('Number of Participants')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```



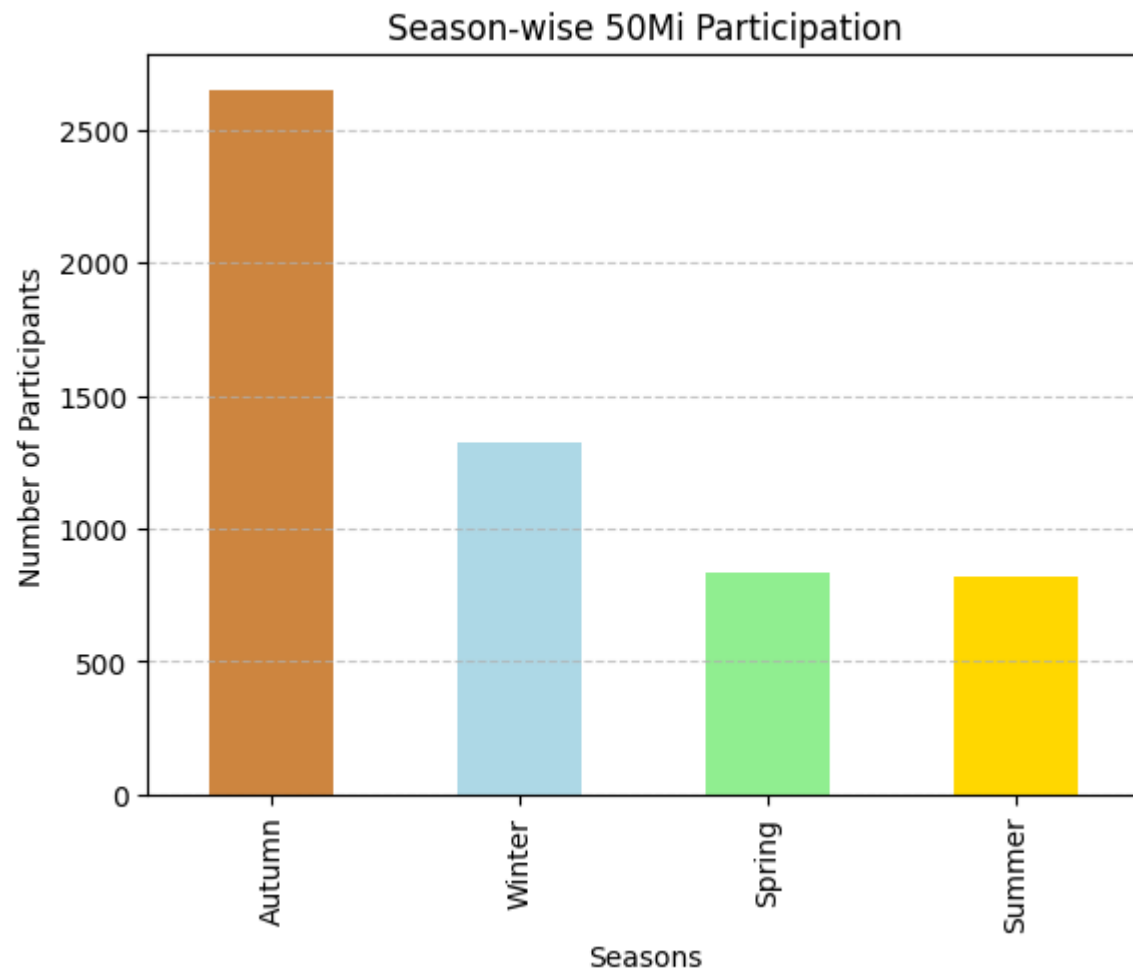
50mi vs 50km seaason-wise participation

50mi Participants


```
In [57]: # 50mi season-wise participation
only_fifty_miles = df3[df3['race_length'] == '50mi']
season_counts_50mi = only_fifty_miles['race_season'].value_counts()
season_counts_50mi
```

```
Out[57]: race_season
Autumn    2647
Winter    1327
Spring     836
Summer     817
Name: count, dtype: int64
```

```
In [58]: season_counts_50mi.plot(kind='bar', color=['#CD853F', '#ADD8E6', '#90EE90', '#FFD700'])
plt.title('Season-wise 50Mi Participation')
plt.xlabel('Seasons')
plt.ylabel('Number of Participants')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```

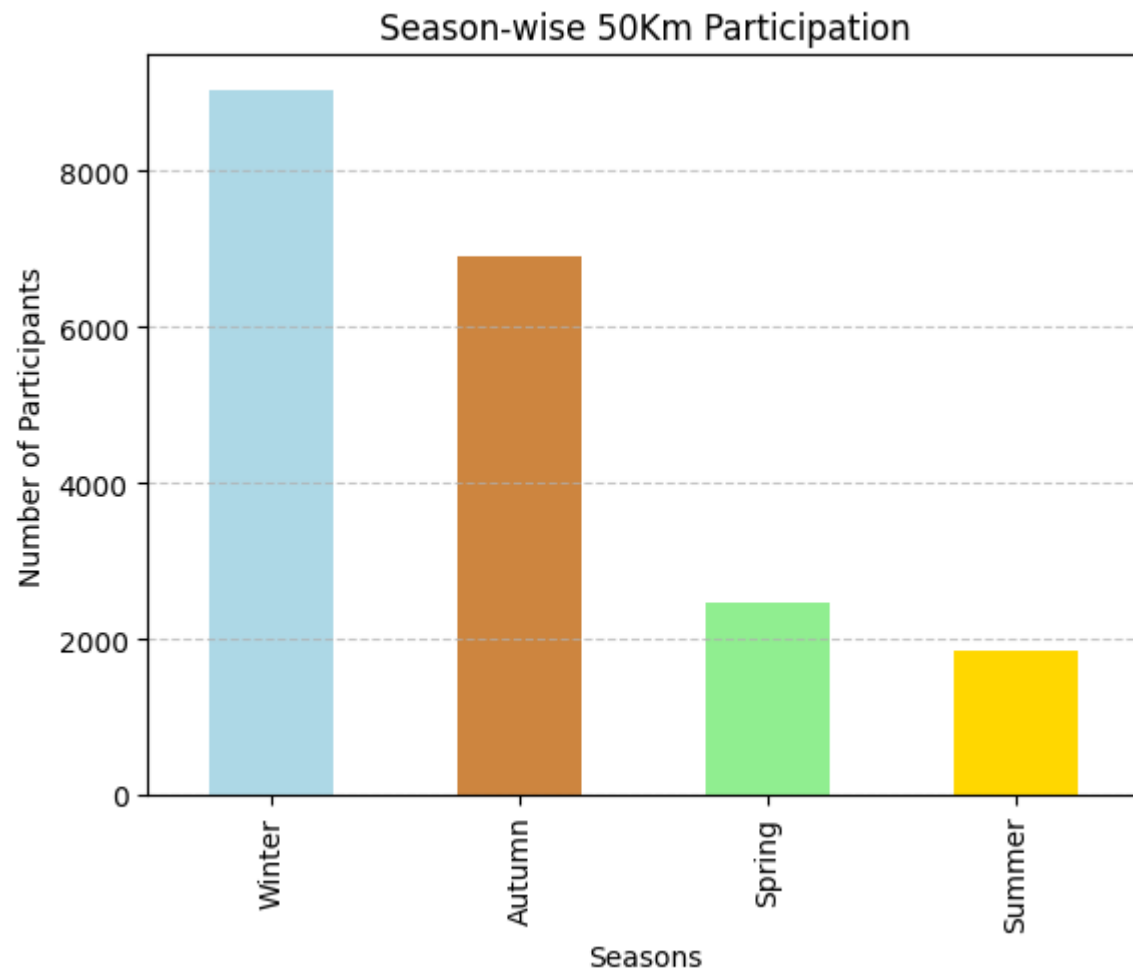


50km Participants

```
In [59]: # 50km season-wise participation
only_fifty_kilometers = df3[df3['race_length'] == '50km']
season_counts_50km = only_fifty_kilometers['race_season'].value_counts()
season_counts_50km
```

```
Out[59]: race_season
        Winter      9034
        Autumn      6902
        Spring       2458
        Summer       1836
        Name: count, dtype: int64
```

```
In [60]: season_counts_50km.plot(kind='bar', color=['#ADD8E6', '#CD853F', '#90EE90', '#FFD700'])
        plt.title('Season-wise 50Km Participation')
        plt.xlabel('Seasons')
        plt.ylabel('Number of Participants')
        plt.grid(axis='y', linestyle='--', alpha=0.7)
        plt.show()
```



In []: